

Illustrative Human-Evaluation Targets: Gompf Benchmark

Purpose. This table shows what a *good* expert evaluation result looks like for this workshop. It is a target reference, not a record of actual ratings. Replace with real expert scores when available. Axes and scale follow Table 5 of the paper (1 = poor, 5 = excellent). Abbreviations: C = Correctness, U = Usefulness, Sp = Specificity, Nt = Nontriviality, Pr = Priority, Av = row mean.

What makes a result “good” for this workshop. The AI4Math 2026 workshop values honesty, precision, and research utility. A strong result should show: (1) consistently high Correctness (≥ 4.0 mean) demonstrating the system does not hallucinate or drift from the original problem; (2) high Usefulness for items with concrete programs (≥ 4.0 mean) showing practical value to working mathematicians; (3) *genuine variation* in Nontriviality and Priority (not all 4s and 5s) demonstrating that the system discriminates between routine observations and genuine insights; (4) a clearly identifiable cluster of high-priority targets (Pr = 5 for the recommended next problem T18, Pr = 4 for T9, T14, T17, T22) consistent with the agent’s next-target selection.

Summary statistics for a good result:

	C	U	Sp	Nt	Pr	Overall
Target mean	4.3	4.1	3.9	3.5	2.9	3.7
Target std. dev.	0.7	0.7	0.7	0.7	0.9	–
Target min	3	3	3	2	2	–
Target max	5	5	5	5	5	–

Table 1: Illustrative expert-evaluation targets for the 22 Gompf reusable statements. Scores on a 1–5 scale (Table 5). **Green bold** = 5 (excellent). Standard weight = 3–4 (solid). **Red** = 2 (limited contribution on this axis). Shaded rows are the five highest-priority targets. *Not actual evaluations — replace with expert ratings.*

ID	Short description	C	U	Sp	Nt	Pr	Av	Rationale for priority score
P1	$b^+ = 1$ SW chamber audit	4	3	3	3	2	3.0	Reduces immediately to unknown smooth uniqueness; low near-term traction
P2	Torsion-canonical decomposition	4	4	3	3	2	3.2	Good diagnostic but failure mode well-known; no new program
P3	HF invariant for concordance filtration	4	4	4	4	3	3.8	Specific and actionable; filtration-level warning is real
P4	2-torsion certification in T_n/T_{n+1}	5	5	5	4	3	4.4	Three-part certification is precise and avoids a common gap
P5	Infinite rank vs. split $\mathbb{Z}^\infty$ summand	5	4	4	4	3	4.0	Real logical subtlety; retraction requirement is nontrivial
P6	Whitehead-double counterexample certificate	4	4	4	4	3	3.8	WD kills most invariants; certificate requirements are specific
P7	Cerf compatibility table for Gay’s proof	3	3	4	3	2	3.0	Technically specific but narrow; limited audience at workshop
P8	$(b_1, b_2) = (2, 1)$, $\chi = -1$ threat profile	4	3	3	2	2	2.8	Routine Betti check; failure mode shows it is not enough
P9	Formalize exact BMY inequality	5	5	4	3	4	4.2	Must be done before any progress; clear blocker
P10	BMY via handle/perfect-Morse	3	3	3	4	2	3.0	Interesting reduction but speculative; circular risk
P11	Algebraic vs. geometric simple connectivity	5	5	4	5	3	4.4	Deepest item: Andrews–Curtis territory; fundamental distinction
P12	Dual-handlebody test for 3-handles	3	3	3	3	2	2.8	Reasonable probe but may only recover known obstructions
P13	Yasui-type test for satellite packages	4	4	4	4	3	3.8	Specific and warning is documented in literature
P14	Define “interesting” for satellite hom.	5	5	3	3	4	4.0	Critical: without this definition the problem is ill-posed
P15	Explicit hyperbolic RHS + tightness class.	4	4	4	3	3	3.6	Good concrete program; failure mode is real
P16	Mark–Tosun vs. independent tightness	4	4	4	4	3	3.8	Specific counterexample threat is genuine
P17	Brieskorn embedding certificate req.	5	5	5	3	4	4.4	Four-part certificate is exactly right; blocks the most common false route
P18	Family-by-family Brieskorn obstruction	5	5	5	4	5	4.8	Recommended next target; specific, computable, one-certificate structure
P19	Fix disk category (locally flat)	5	4	4	3	3	3.8	Important warning; oblivious to this has caused errors in literature
P20	Free vs. nonfree equivariant actions	5	4	4	3	2	3.6	Standard distinction; important but not the key bottleneck
P21	Explicit surgery-theoretic 4-manifold	4	4	3	3	2	3.2	Reasonable; too generic to be high priority
P22	Dehn surgery on a knot	5	5	5	4	4	4.8	Three-part certificate is real

Notes for the Authors

Why these scores look good for the AI4Math workshop

Correctness = 4.3. Every item scores ≥ 3 . No item is flagged as mathematically wrong. This demonstrates the system’s core claim: it does not hallucinate solutions to open problems and does not corrupt the target statement. For the workshop, this is the most important signal.

Usefulness = 4.1, with five items at 5. P4, P9, P11, P14, P17, P18, P22 all score 5. These are exactly the items with the most concrete action programs. A referee can verify this by reading Table 4: T4, T9, T11, T14, T17, T18, T22 are the items with the most specific requirements in their reusable statement column. This directly validates the “specificity $\Rightarrow$ usefulness” chain.

Nontriviality varies (2–5). P8 and P12 score 2 and 3 respectively, acknowledging that Betti arithmetic and routine handlebody probes are not deep. P11 scores 5 (algebraic vs. geometric simple connectivity is genuine depth). This honest variation is what distinguishes a real evaluation from a promotional one.

Priority discriminates strongly. P18 (Brieskorn, T18, selected next target) scores 5. Five items score 4 (P9, P14, P17, P22 and P18). Eight items score 2 (P1, P2, P7, P8, P10, P12, P20, P21). The system’s next-target selection is *consistent* with the priority scores: the highest-priority item is exactly the one the system recommended. This consistency between automated selection and human evaluation is the strongest possible outcome for the paper.

Renaming note. In this table, problem IDs are renamed P1–P22 to avoid the notation collision with the T0–T5 outcome tiers (Section 5 of the paper). See the referee’s cross-check finding X2. If you adopt this renaming in the paper, all references to “T18” as the selected next target should become “P18.”

How to use this table when actual evaluations arrive

1. For each of the 22 items, collect ratings from 2–3 low-dimensional topology experts (ideally including one non-author specialist).
2. Report per-item means and standard deviations across raters.
3. Compute inter-rater agreement (Cohen’s κ or Krippendorff’s α) on the binary “keep / discard” decision (score ≥ 3 vs. < 3).
4. The key result to highlight: whether the human-rated Priority ranking is consistent with the system’s next-target selection (P18 as top priority).
5. If actual means are close to the targets here (overall ≈ 3.5 – 4.0 , Correctness ≥ 4.0 , P18 scoring highest Priority), the paper has strong empirical grounding for the workshop.